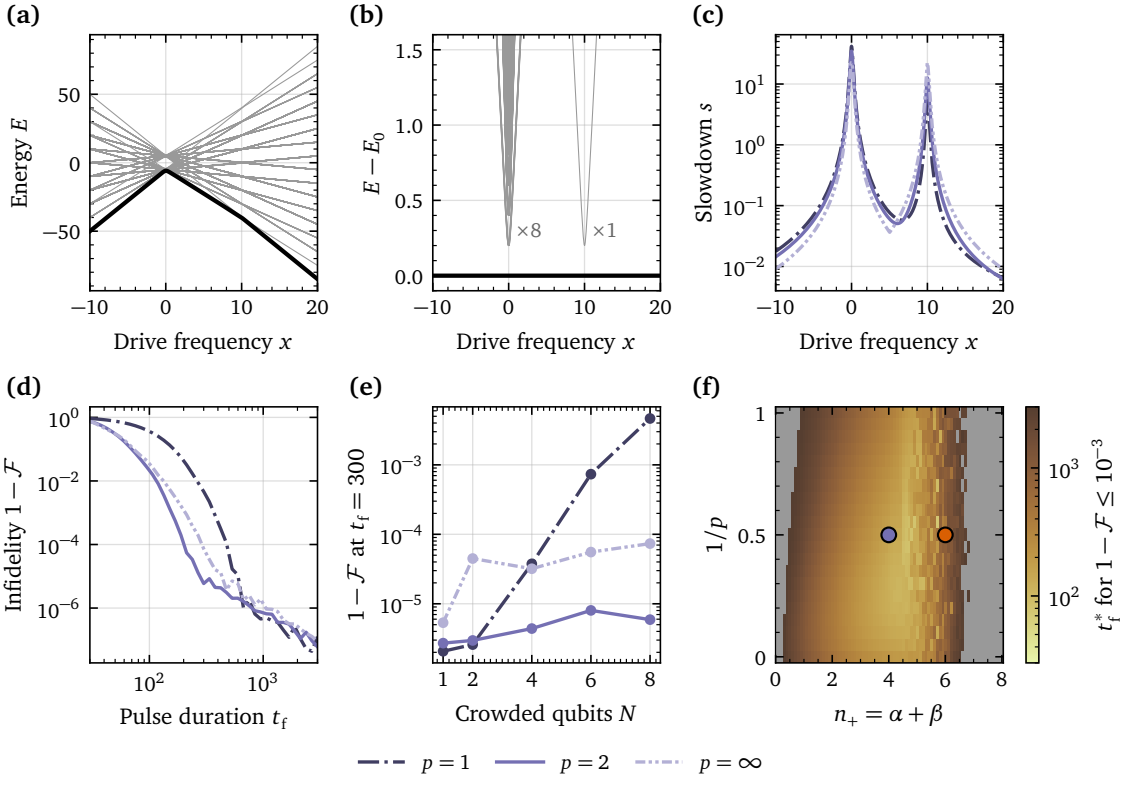

(a)
Energy E
50
0
−50
−10
0
10
20
Drive frequency x
(b)
$E - E_0$
1.5
1.0
0.5
0.0
×8
×1
−10
0
10
20
Drive frequency x
(c)
Slowdown s
10^{1}
10^{0}
10^{-1}
10^{-2}
−10
0
10
20
Drive frequency x
(d)
Infidelity $1 - \mathcal{F}$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{2}
10^{3}
Pulse duration t_f
(e)
$1 - \mathcal{F}$ at $t_\mathrm{f} = 300$
10^{-3}
10^{-4}
10^{-5}
1
2
4
6
8
Crowded qubits N
(f)
$1/p$
1
0.5
0
0
2
4
6
8
$n_+ = \alpha + \beta$
10^{3}
10^{2}
t_f^* for $1 - \mathcal{F} \leq 10^{-3}$
$p = 1$
$p = 2$
$p = \infty$